MTH 330 Term Project Checkpoint

Balancing Market Impact and Price Risk: Choosing a Trade Execution Horizon

1. Group Members

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2. Project Topic

Tentative title: Balancing Market Impact and Price Risk: Choosing a Trade Execution Horizon

Problem. A trading program must sell Q shares within a fixed time. Selling fast pushes the price down, an effect called market impact. Selling slowly leaves the unsold shares exposed to price swings, called timing risk. Our project asks how long the sale should take if the program sells at an even pace.

Scope. The full problem asks how selling speed should change from moment to moment. Allowing any pattern of changing speeds generally needs methods beyond single-variable calculus. We keep the selling rate constant and choose only the duration. A simple two-stage pattern could be a later extension, but it is not part of this model.

Relevance. Trade execution is a computing problem. Large orders are often split into smaller trades and sent by software, and that software needs a rule for how fast to sell. One standard schedule, the time-weighted average price (TWAP) strategy, sells at an even pace, which matches our model. The trade-off also makes a clean calculus question: one cost falls as the sale gets longer and the other rises, so a best duration can sit between them. Research by Almgren and Chriss guides our assumptions for the price effect and the uncertainty. We study a simplified version and do not reproduce their full solution.

3. Research Direction

What we will model. Let Q be the number of shares to sell, T the total sale duration in minutes, and t the time elapsed. With an even pace, v is the selling rate and $x(t)$ is the number of shares still unsold. For $0 \le t \le T$:

$$v = \frac{Q}{T}, \qquad x(t) = Q\left(1 - \frac{t}{T}\right)$$

Price effect. We assume the price reduction per share is proportional to selling speed, with positive coefficient η. Selling v shares per minute then gives a cost rate of ηv^2 dollars per minute, so faster selling raises the cost rate.

Uncertainty. Let σ be arithmetic price volatility: the scale of unpredictable changes in dollar price per share, not in percentage returns. Let λ be a positive weight for how strongly we want to avoid uncertainty. The uncertainty-penalty rate is $\lambda\sigma^2 x(t)^2$, since more unsold shares leave more money exposed to price changes. This term penalizes variance, the spread of outcomes around the expected result. It is not an expected loss.

Combined score. Integrals accumulate each rate over the sale into one cost-and-risk score $J(T)$, in dollars:

$$J(T) = \int_0^T \eta\, v^2\, dt + \lambda\sigma^2 \int_0^T x\,(t)^2\, dt$$

Our next step is to derive and verify

$$J(T) = \frac{\eta Q^2}{T} + \frac{\lambda\sigma^2 Q^2 T}{3}$$

within this even-pace model. The first term falls as T grows and the second rises.

Main questions.

1. Which allowed duration minimizes $J(T)$?
2. How does that duration change with σ?
3. How does it change with λ?

For questions 2 and 3, we vary one parameter at a time and hold Q and η fixed.

Inputs and assumptions. We will set a shortest duration $T_{\min}$ and a longest duration $T_{\max}$, with $0 < T_{\min} < T_{\max}$, and search between them, endpoints included. All inputs will be positive illustrative values, not values fitted to market data. Closing prices alone cannot determine η or λ. With Q in shares and T in minutes, the units are σ in dollars/$(\text{share} \cdot \sqrt{\text{minute}})$, η in dollars $\cdot$ minute/share^2, and λ in 1/dollar.

We assume trading conditions and volatility stay unchanged, the price has no upward or downward trend, the sale leaves no lasting price effect, and every share is sold with no buying back. We do not model reactions to changing prices or limits based on overall market trading volume.

4. Proposed Mathematical and Python Methods

Integration (CLO 6–7). We will integrate the two rates and interpret each result: the total expected impact cost, and the accumulated exposure from squared unsold shares.

Limits and continuity. Limits describe J as $T \to 0^+$ and as $T \to \infty$, where we expect it to grow without bound. Continuity on the closed interval $[T_{\min}, T_{\max}]$ guarantees that a minimum exists, by the Extreme Value Theorem. We never evaluate $T = 0$.

Derivatives and optimization. The first derivative $J'(T)$ gives candidate durations. The second derivative $J''(T)$ shows whether the unrestricted positive-duration minimum is unique. We then compare the allowed candidates with the endpoints.

Sensitivity. Changing σ and λ separately answers the remaining two questions. Any best duration applies only to a constant selling rate, not to every possible pattern of sales.

Python. One Jupyter notebook will hold all the work:

- **SymPy** for exact integration, derivatives and limits.
- **NumPy** for numerical comparisons across a grid of durations.
- **Matplotlib** for graphs of both cost components, shares remaining over time, and changes in the preferred duration.
- **SciPy** (optional) to check the integrals and search for the minimum.

We will compare numerical answers with the symbolic expressions and endpoint checks, record every input, and label each graph with units.

These are proposed steps, not completed calculations or code, and the methods may change. Under this model's scaling, Q cancels from the preferred duration. Our discussion will explain why, without concluding that order size never matters in practice.

5. Plan for Meeting the Project Rubric

Criterion (weight)	Our plan
Problem Definition & Relevance (10%)	Open with the sell-fast-or-slow dilemma and its link to execution software such as TWAP schedules.
Mathematical Modeling (15%)	Justify each function, assumption and unit. Keep the expected price cost separate from the uncertainty penalty.
Use of Calculus Concepts, CLO 1–5 (20%)	Apply limits, continuity, derivatives and optimization, with the reasoning written out for each step.
Integration & Area Analysis, CLO 6–7 (10%)	Interpret both integrals in context: accumulated impact cost and accumulated squared-unsold-shares exposure.
Python Implementation (15%)	One documented, reproducible notebook, with symbolic results checked against numerical ones.
Visualization & Interpretation (10%)	Labeled graphs with units, each followed by a short explanation of what the comparison shows.
Analytical Insight & Discussion (10%)	Interpret how σ and λ shift the best duration, and state where the simple model may fail.
Report Organization & Presentation (5%)	Follow the path from question to model to method, with the same notation throughout.
Creativity and Originality (5%)	Our own explanations and illustrative comparisons, with credit to the underlying research.

Key Terms

Trade execution: carrying out a purchase or sale; liquidation here means selling all the chosen shares. **Horizon:** the duration allowed for the sale. **Inventory:** shares still unsold, $x(t)$. **Market impact:** a price change caused by the trade itself; temporary impact is the immediate disadvantage during selling, and linear impact means the per-share disadvantage is proportional to selling speed. **Arithmetic price volatility (σ):** the scale of random changes in dollar price per share, per square root of a minute. **Variance:** expected squared deviation from the expected outcome. **Risk aversion (λ):** preference for avoiding uncertainty; λ times variance is a decision penalty, not a predicted loss. **Sensitivity comparison:** changing one input while holding the others fixed.

References

Almgren, R., & Chriss, N. (2000, December). *Optimal execution of portfolio transactions* [Manuscript], §§1.4–1.5, 2.2. https://www.smallake.kr/wp-content/uploads/2016/03/optliq.pdf. Supports the general cost-and-variance framework, not our exact constant-rate horizon reduction.

New York University. (2013). *Risk and portfolio management*, Assignment 2, Problems 1–2. https://math.nyu.edu/inmemoriam/avellaneda/Project2.2013.pdf. Treats time as continuous and holdings in dollars rather than shares; we do not mix its coefficients with ours.